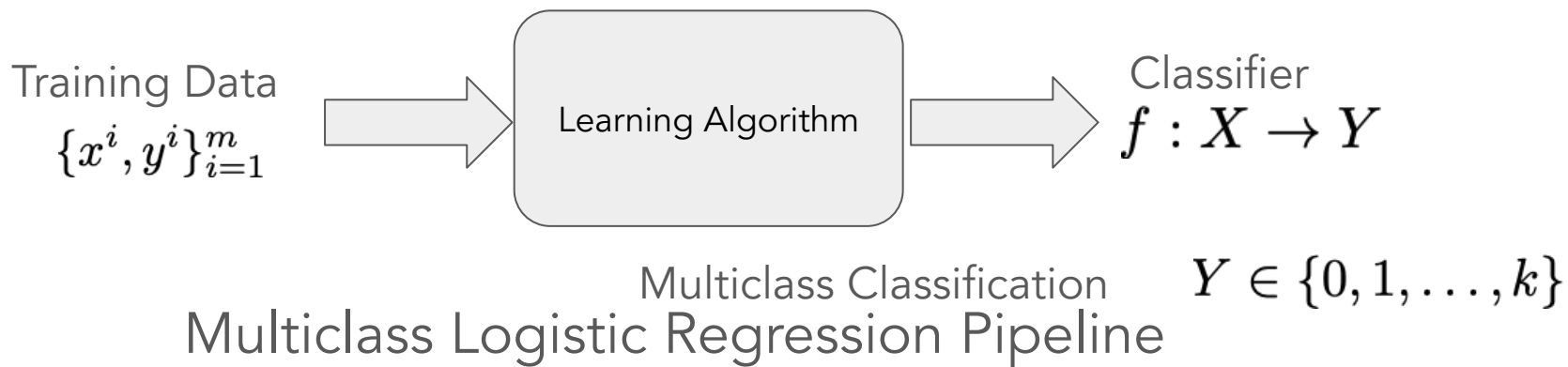


CX4240 Spring 2026

Recurrent Neural Networks

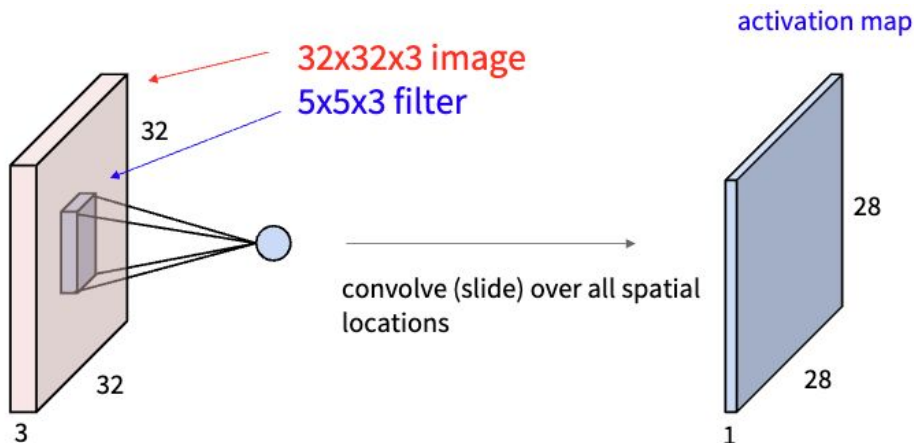
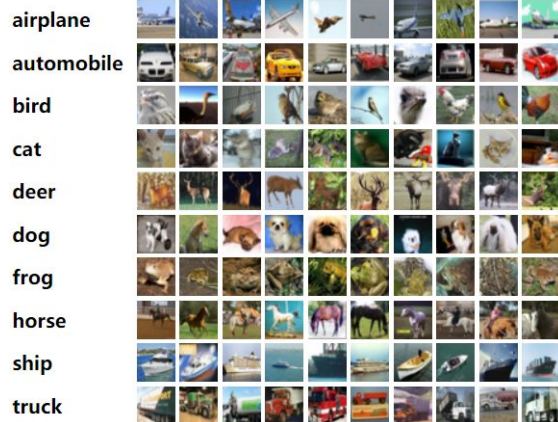
Bo Dai
School of CSE, Georgia Tech
bodai@cc.gatech.edu

Multiclass Logistic Regression Algorithms

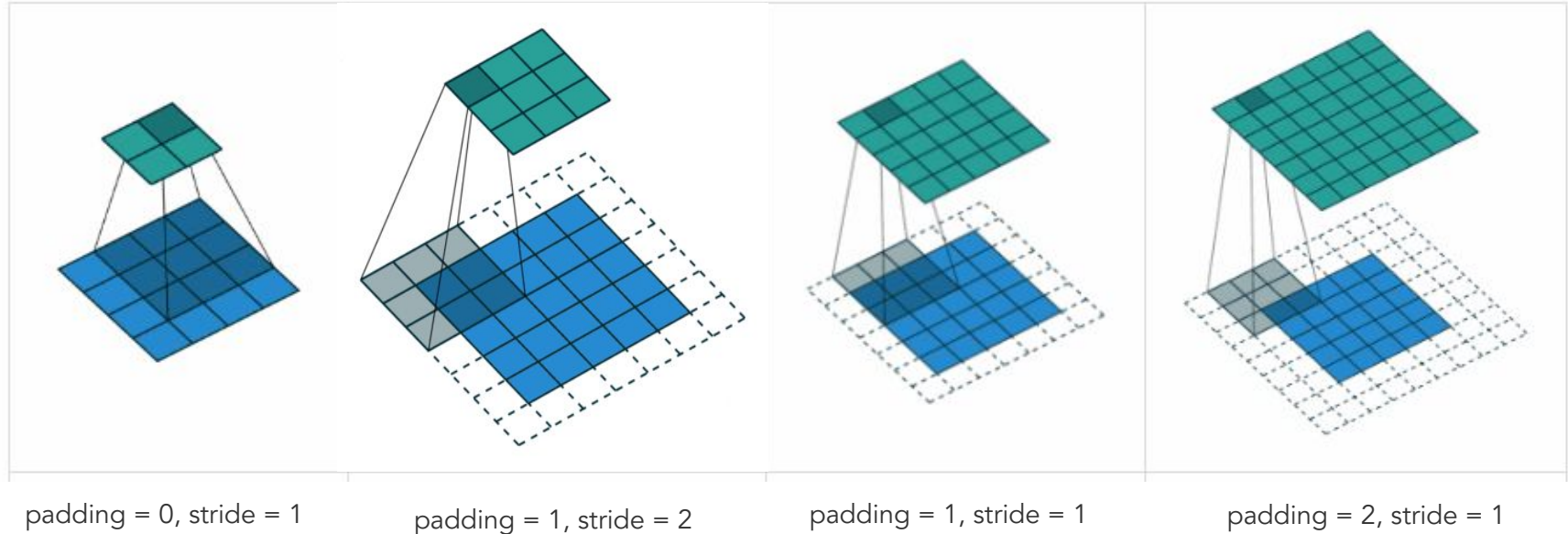


1. Build probabilistic models:
Categorical Distribution + Conv NN
2. Derive loss function: MLE and MAP
3. Select optimizer: (Stochastic) Gradient Descent

Revisit Convolution Neural Networks



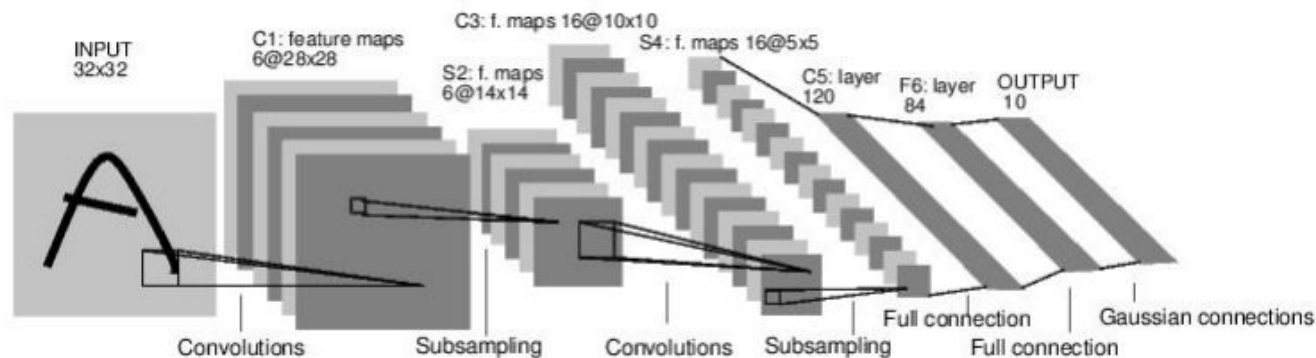
Convolution for 2D Images



$$W_{out} = \frac{W - F + 2P}{S} + 1$$

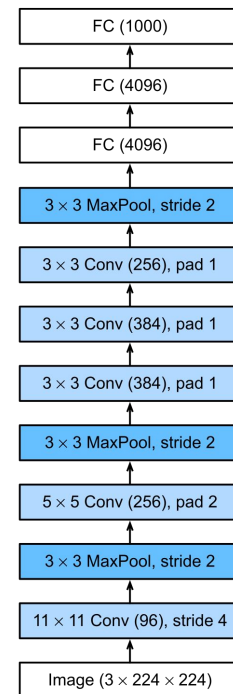
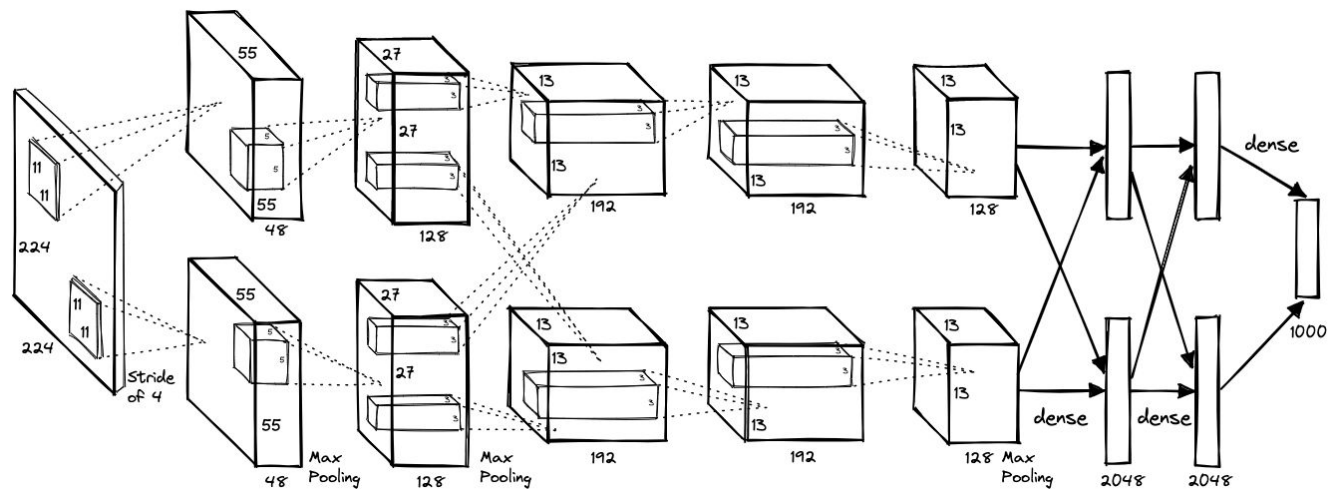
Animation from [Hochschule der Medien](https://www.hochschule-der-medien.de/)

Put Everything Together for Images



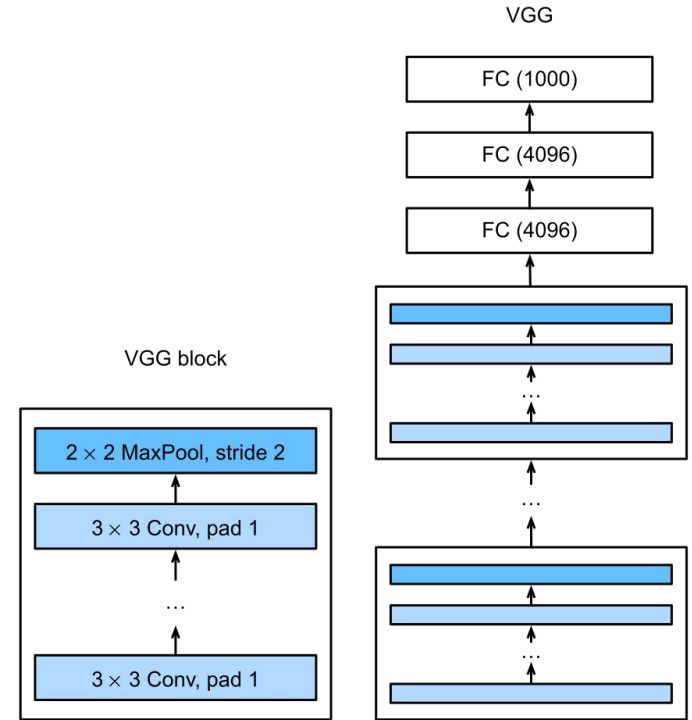
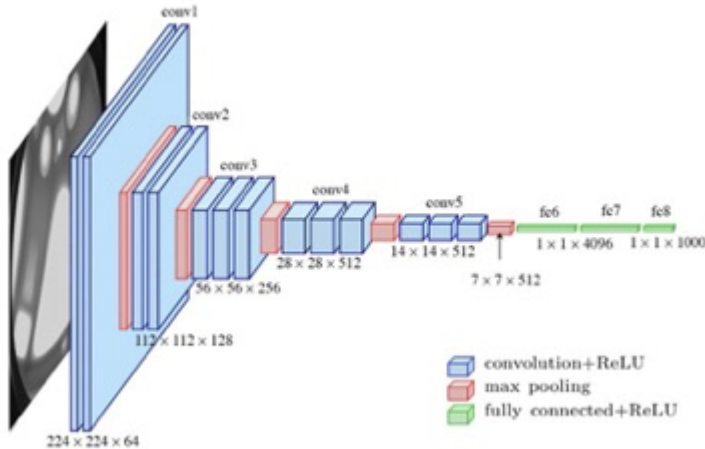
[LeNet-5, LeCun 1980]

AlexNet (2012)



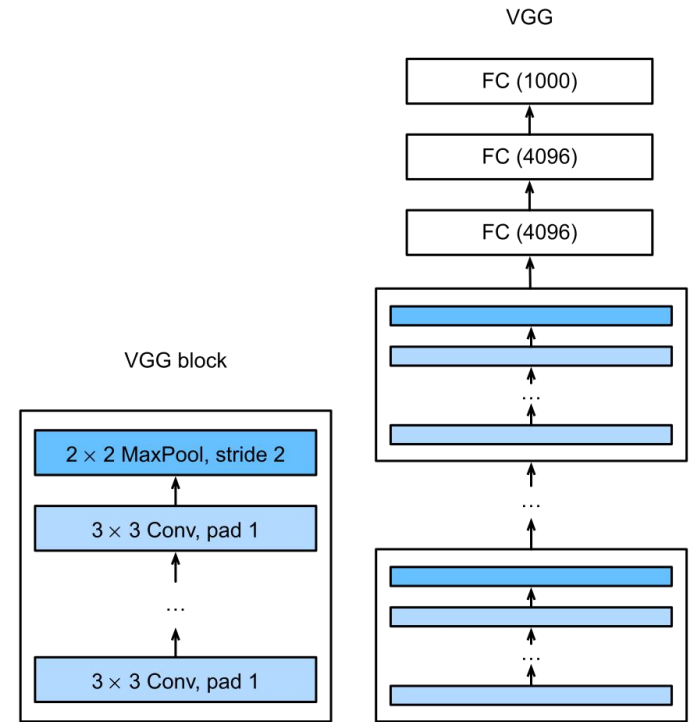
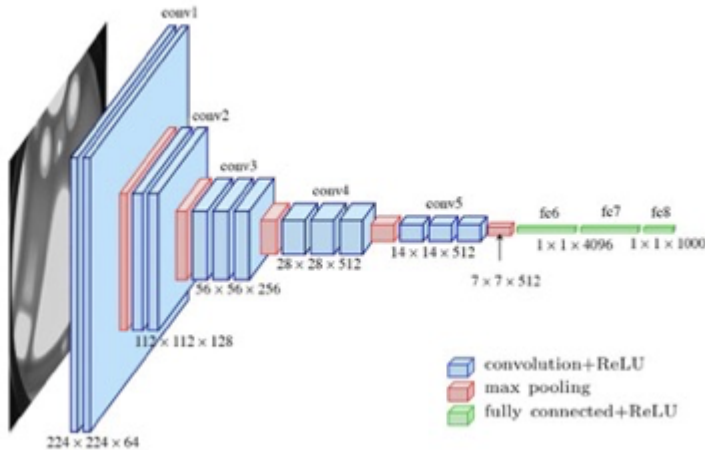
VGGNet (2014)

- Very Deep CNN
- With only 3*3 conv filters
 - Fewer parameters, deeper nonlinear layers



VGGNet (2014)

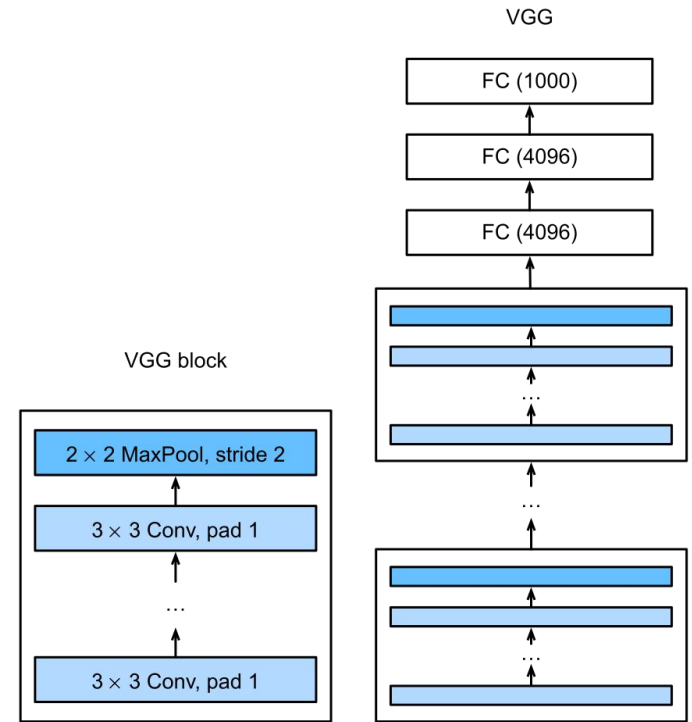
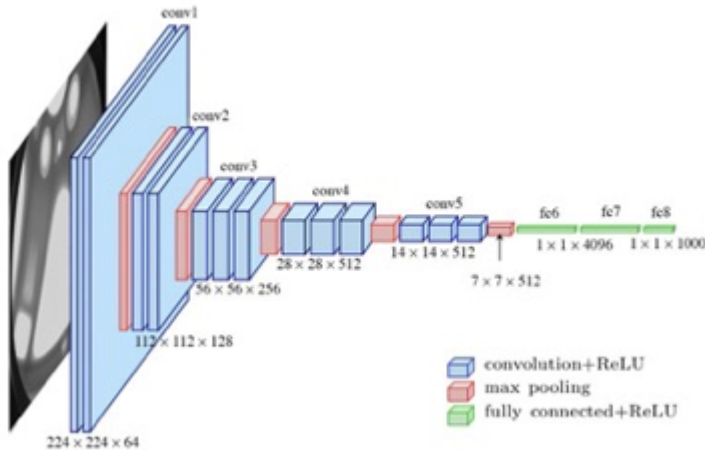
- Very Deep CNN
- With only 3*3 conv filters
 - Fewer parameters, deeper nonlinear layers



Keep increasing the depth?

VGGNet (2014)

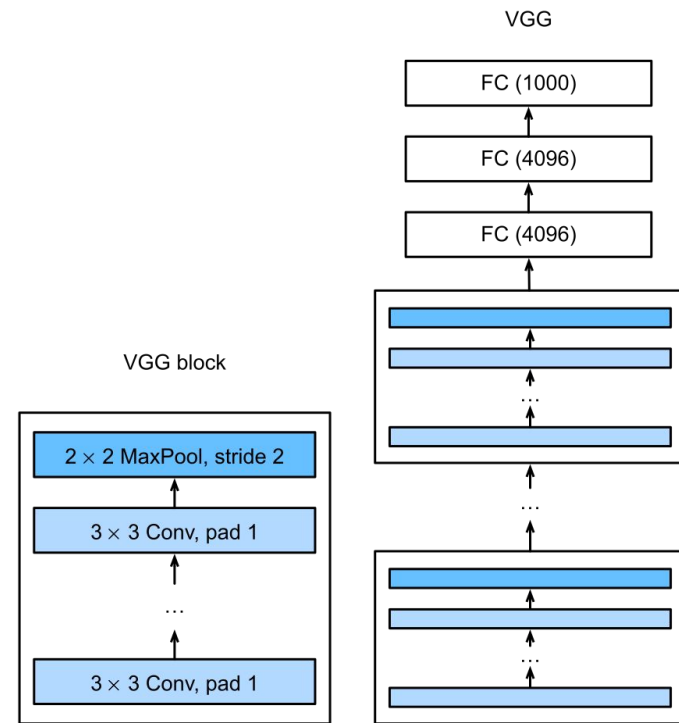
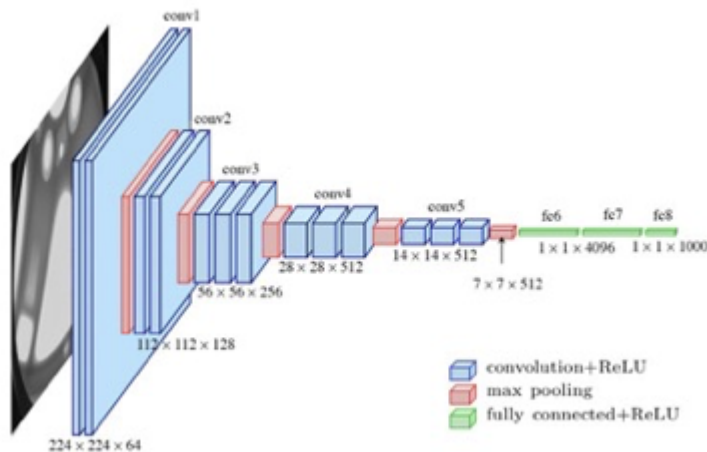
- Very Deep CNN
- With only 3*3 conv filters
 - Fewer parameters, deeper nonlinear layers



vanishing or exploding

VGGNet (2014)

- Very Deep CNN
- With only 3*3 conv filters
 - Fewer parameters, deeper nonlinear layers

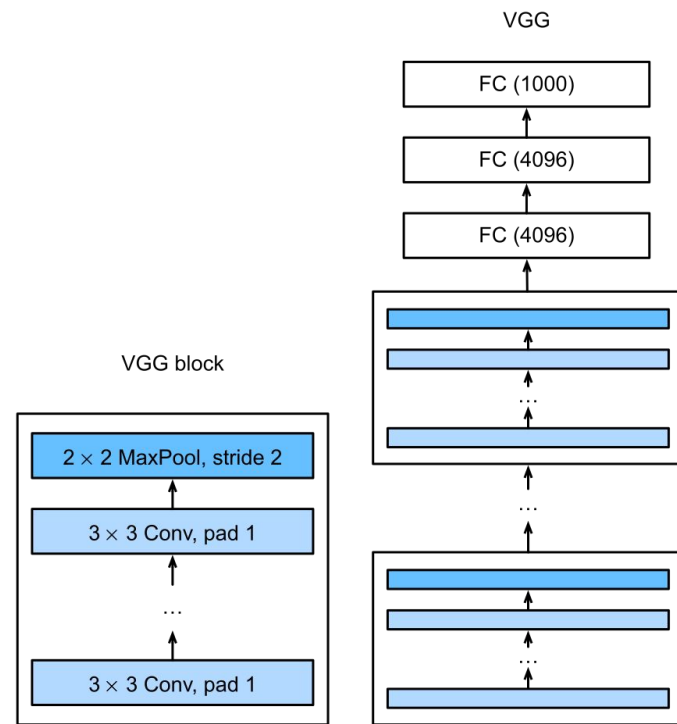
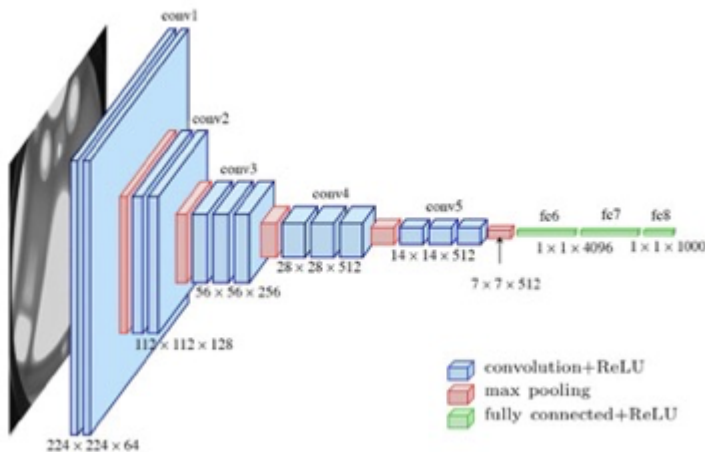


vanishing or exploding

$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial a_n} \cdot \frac{\partial a_n}{\partial a_{n-1}} \cdot \frac{\partial a_{n-1}}{\partial a_{n-2}} \cdot \dots \cdot \frac{\partial a_{i+1}}{\partial a_i} \cdot \frac{\partial a_i}{\partial w_i}$$

VGGNet (2014)

- Very Deep CNN
- With only 3*3 conv filters
 - Fewer parameters, deeper nonlinear layers

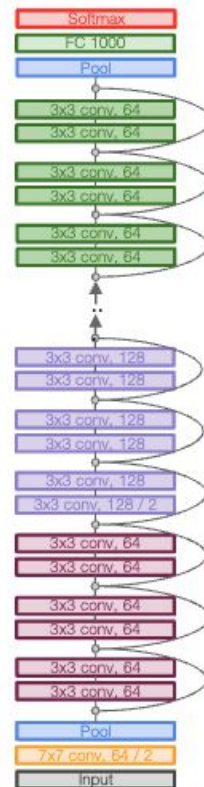
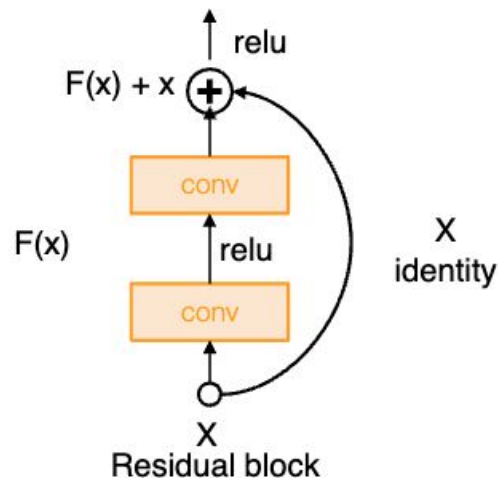


vanishing or exploding

$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial a_n} \cdot \frac{\partial a_n}{\partial a_{n-1}} \cdot \frac{\partial a_{n-1}}{\partial a_{n-2}} \cdots \frac{\partial a_{i+1}}{\partial a_i} \cdot \frac{\partial a_i}{\partial w_i} \quad \frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial a_n} \cdot \prod_{j=i}^n \frac{\partial a_j}{\partial a_{j-1}}$$

ResNet (2015)

- Very Deep CNN with residual connections
 - 152-layer model for ImageNet
 - ILSVRC'15 classification winner (3.57% top 5 error)
 - Swept all classification and detection competitions in ILSVRC'15 and COCO'15!

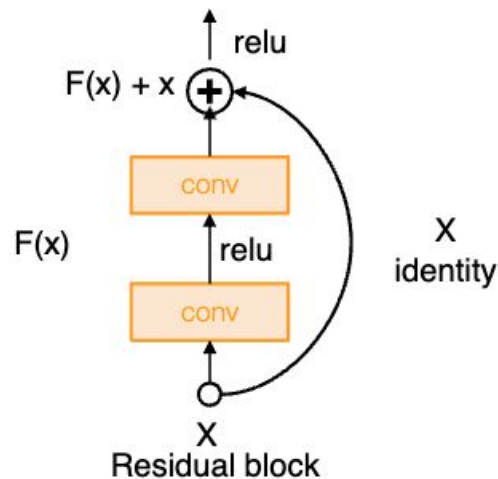


ResNet (2015)

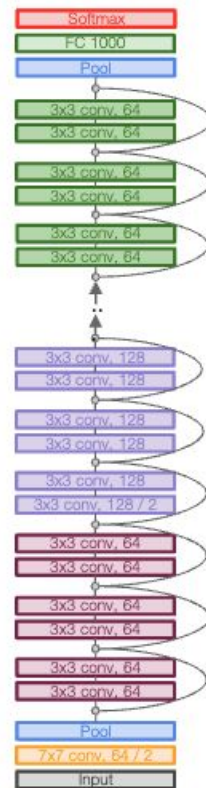
- Very Deep CNN with residual connections
 - **152-layer** model for ImageNet
 - ILSVRC'15 classification winner (3.57% top 5 error)
 - Swept all classification and detection competitions in ILSVRC'15 and COCO'15!
- Migrate gradient exploding and vanishing

$$y = x + F(x)$$

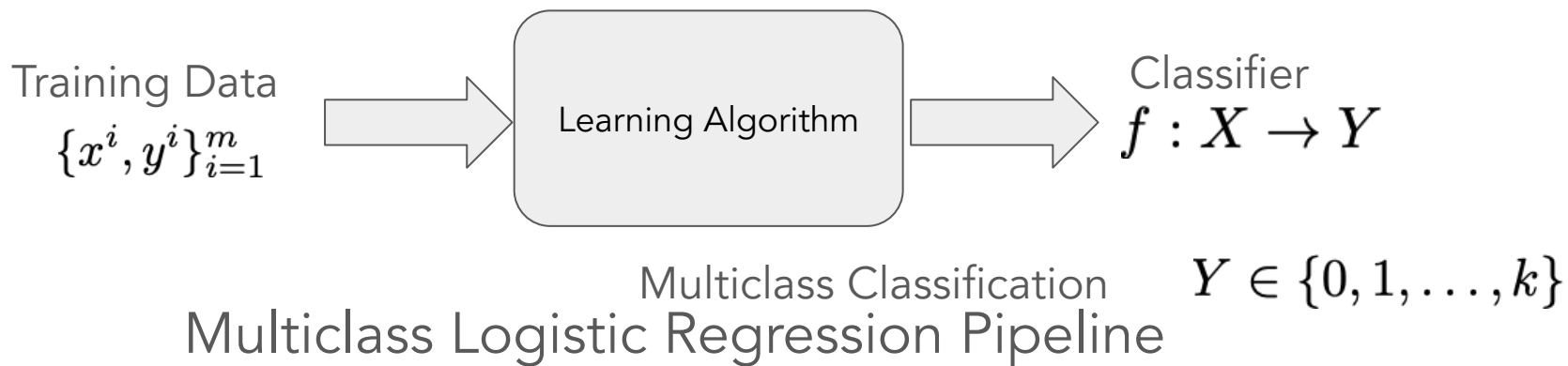
$$\begin{aligned}\frac{\delta E}{\delta x} &= \frac{\delta E}{\delta y} * \frac{\delta y}{\delta x} = \frac{\delta E}{\delta y} * (1 + F'(x)) \\ &= \frac{\delta E}{\delta y} + \frac{\delta E}{\delta y} * F'(x)\end{aligned}$$



$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial a_n} \cdot \prod_{j=i}^n \frac{\partial a_j}{\partial a_{j-1}}$$



Multiclass Logistic Regression Algorithms



1. Build probabilistic models:
Categorical Distribution + Conv NN
2. Derive loss function: MLE and MAP
3. Select optimizer: (Stochastic) Gradient Descent

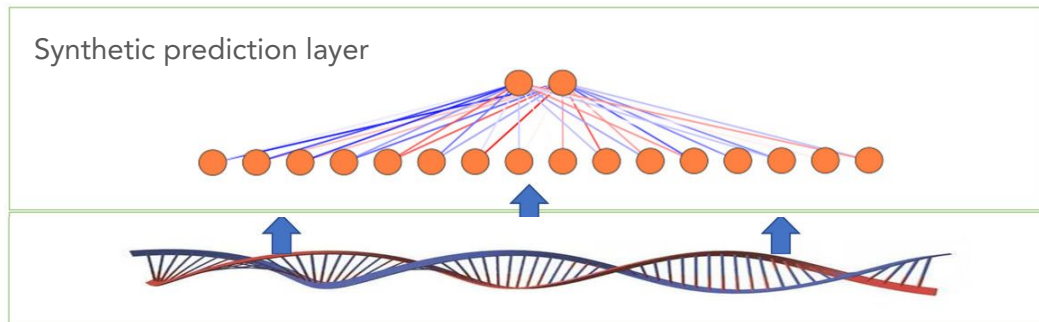
Sequence Prediction



texts[0]

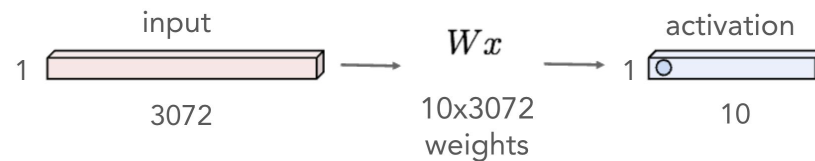
For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

Sequence Prediction

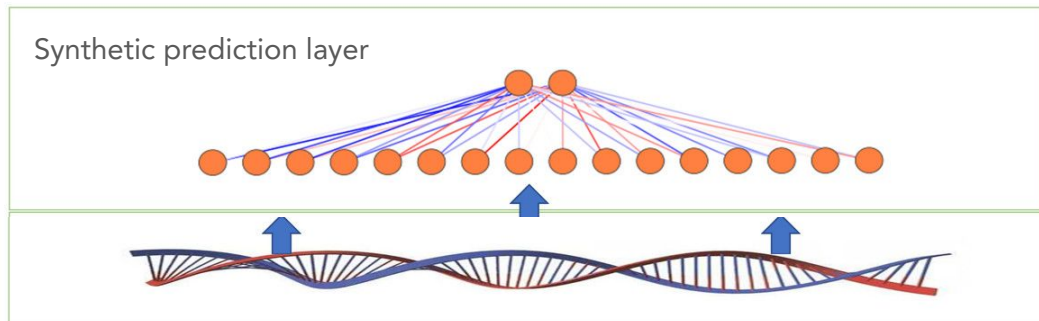


texts[0]

For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

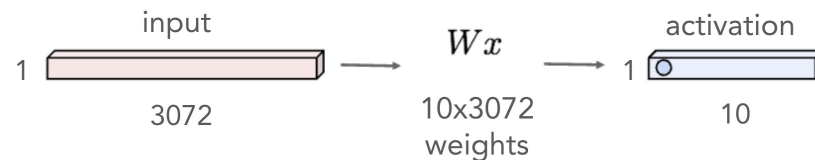


Sequence Prediction



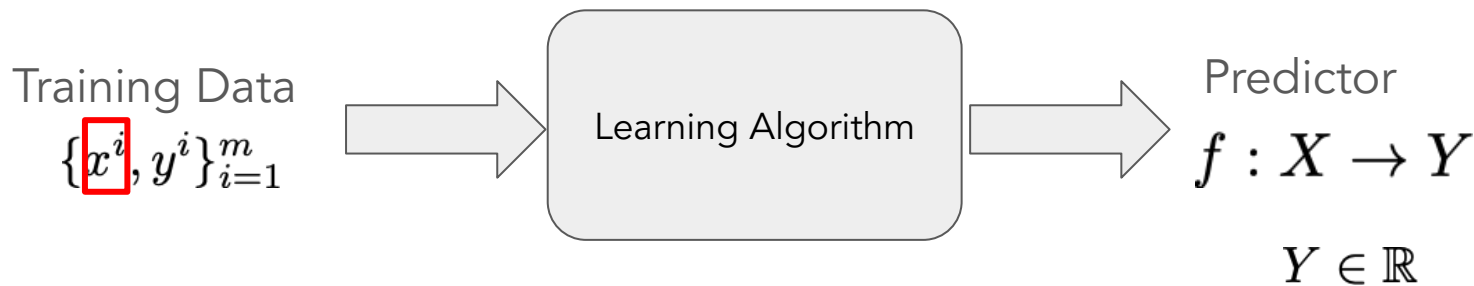
I saw the movie with two grown children. Although it was not as clever as Shrek, I thought it was rather good. In a movie theatre surrounded by children who were on spring break, there was not a sound so I know the children all liked it. There parents also seemed engaged. The death and apparent death of characters brought about the appropriate gasps and comments. Hopefully people realize this movie was made for kids.

For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.



What if the length of sequences varies?

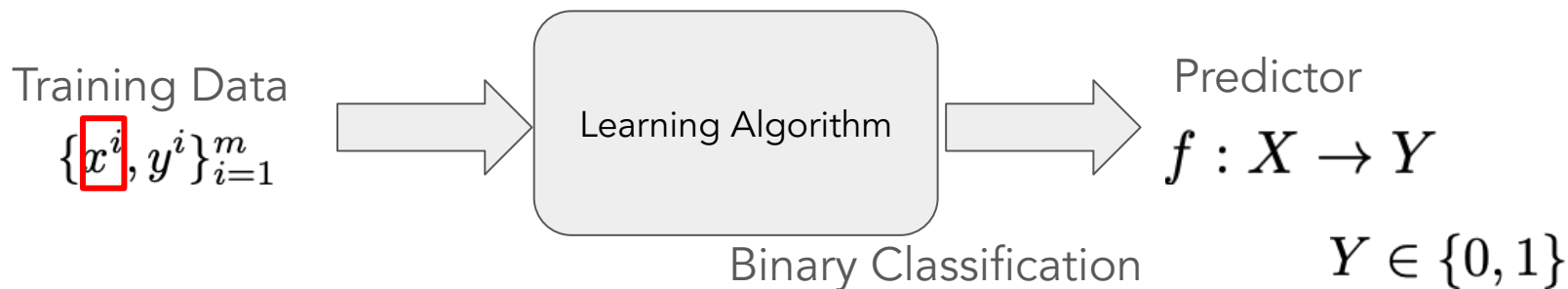
Sequential Regression Algorithms



Linear Regression Pipeline

1. Build probabilistic models:
Gaussian Distribution + RNN
2. Derive loss function: MLE and MAP
3. Select optimizer: (Stochastic) Gradient Descent

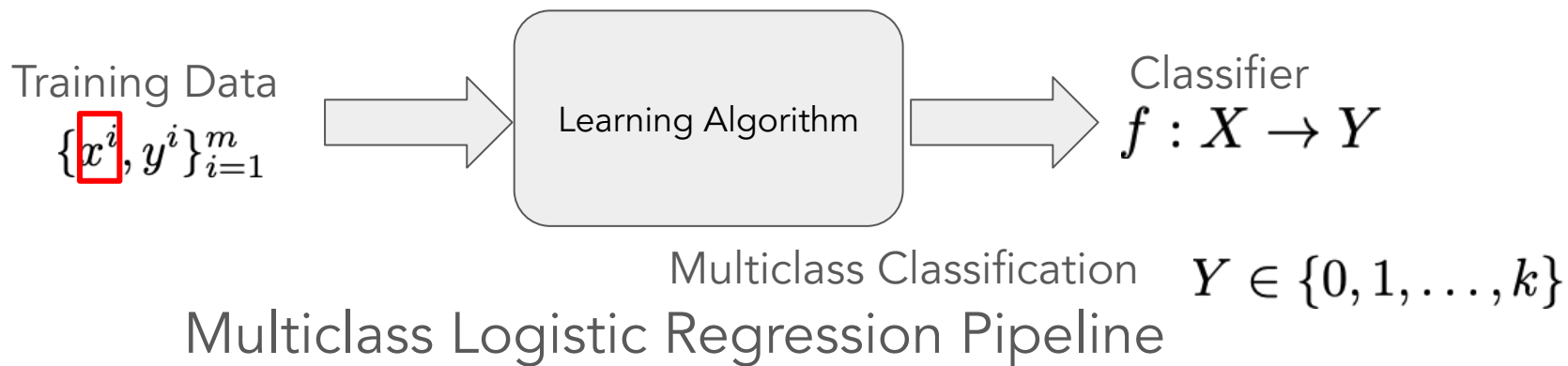
Sequential Binary Classification Algorithms



Binary Logistic Regression Pipeline

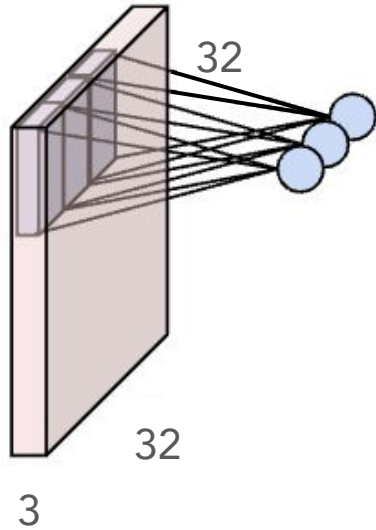
1. Build probabilistic models:
Bernoulli Distribution + RNN
2. Derive loss function: MLE and MAP
3. Select optimizer: (Stochastic) Gradient Descent

Sequential Multiclass Logistic Regression Algorithms



1. Build probabilistic models:
Categorical Distribution + RNN
2. Derive loss function: MLE and MAP
3. Select optimizer: (Stochastic) Gradient Descent

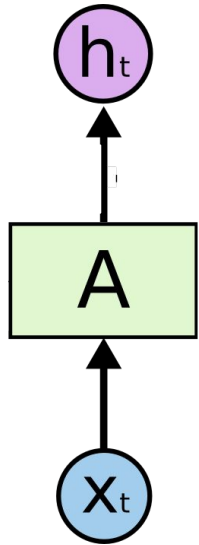
Inspiration from Convolution Layer



one computation cell is shared



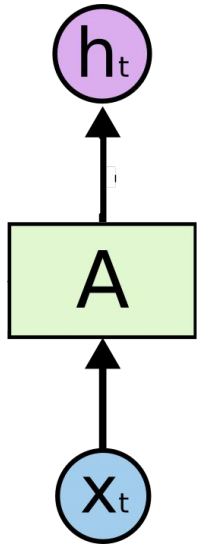
Recurrent Neural Network



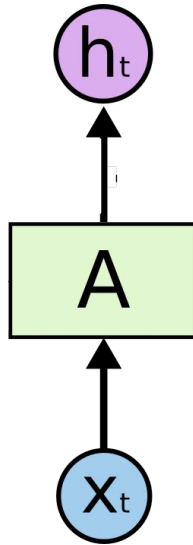
For

For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

Recurrent Neural Network



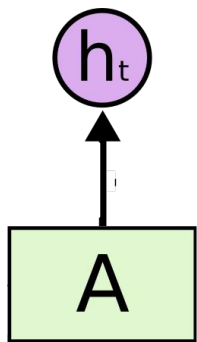
For



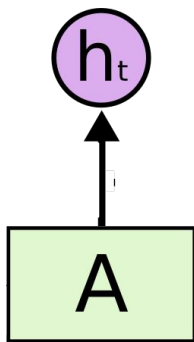
a

For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

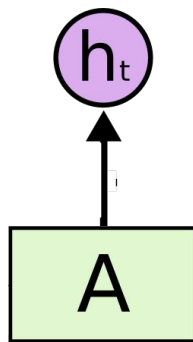
Recurrent Neural Network



For



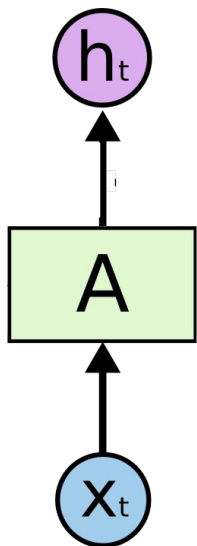
a



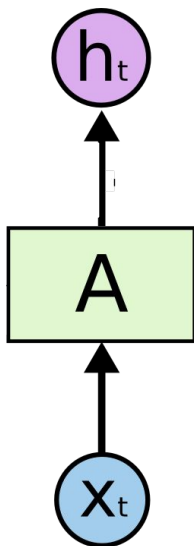
movie

For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

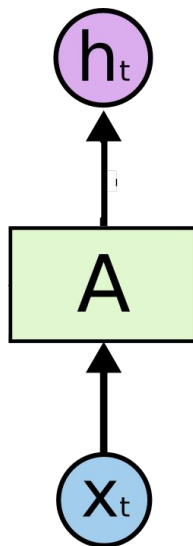
Recurrent Neural Network



For



a

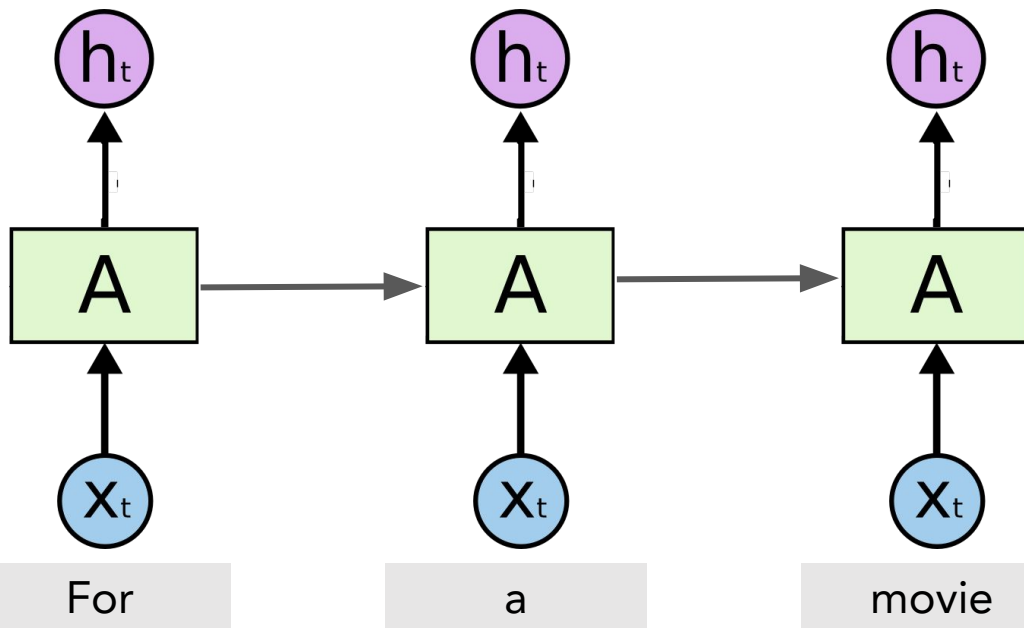


movie

Order Matters!

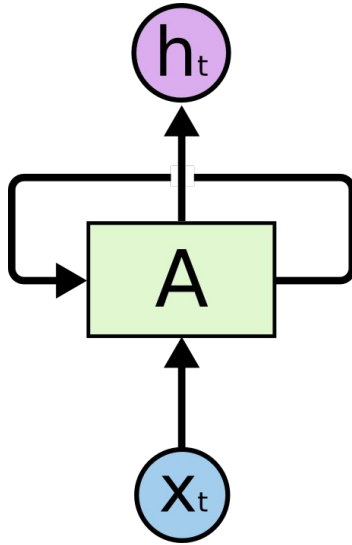
For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

Recurrent Neural Network



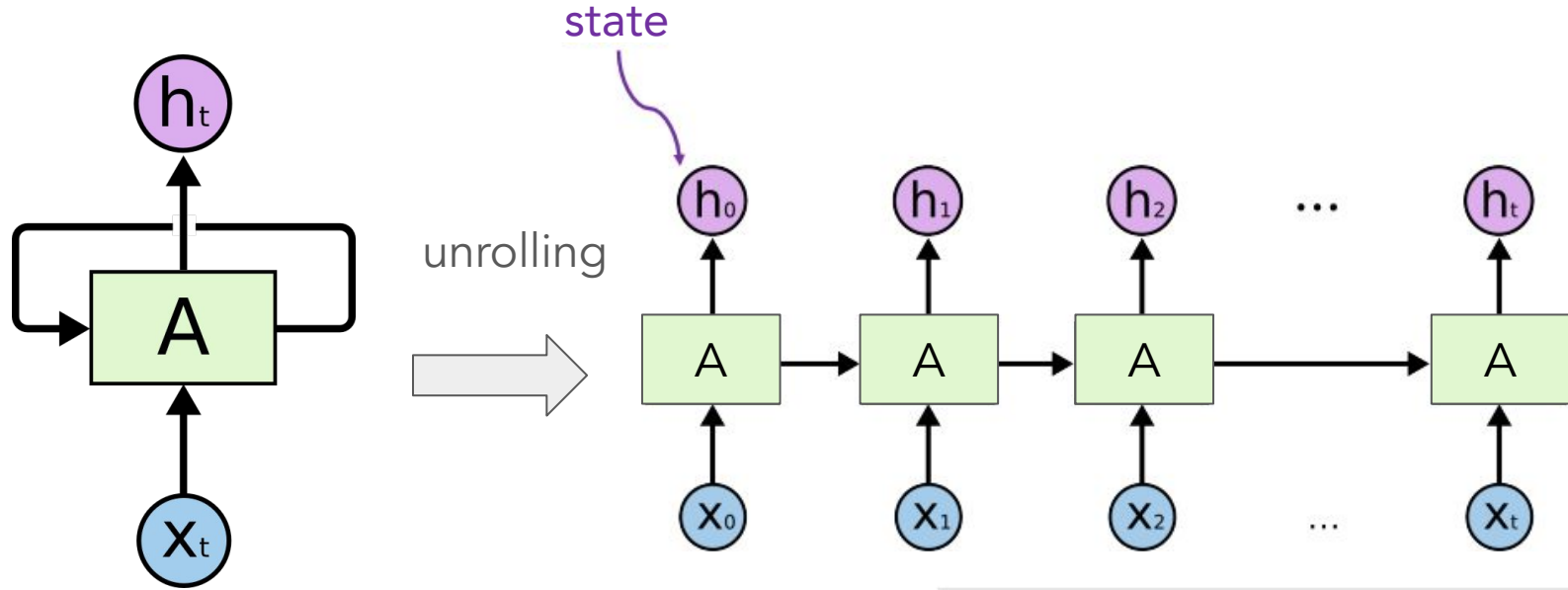
For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

Recurrent Neural Network



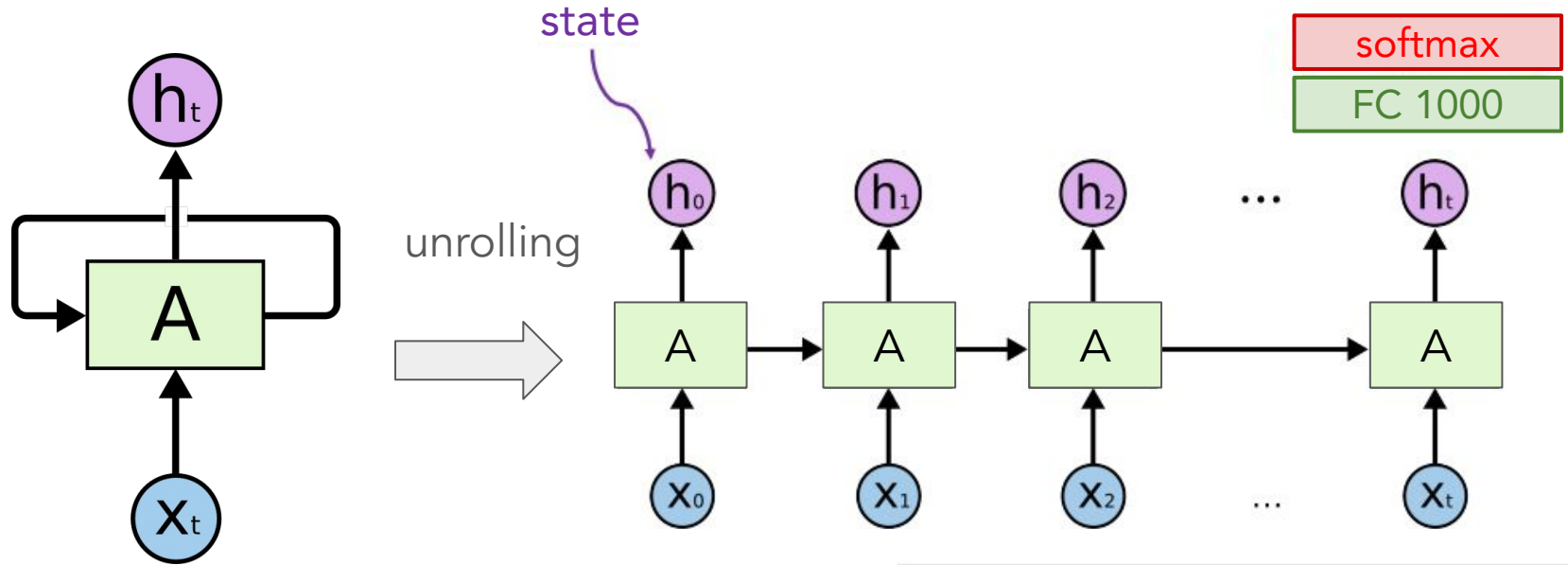
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Recurrent Neural Network



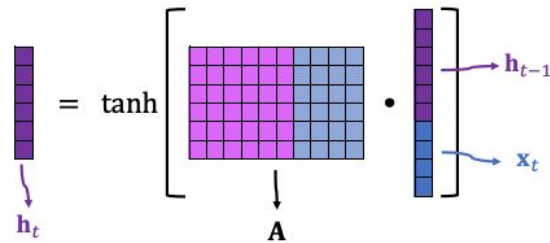
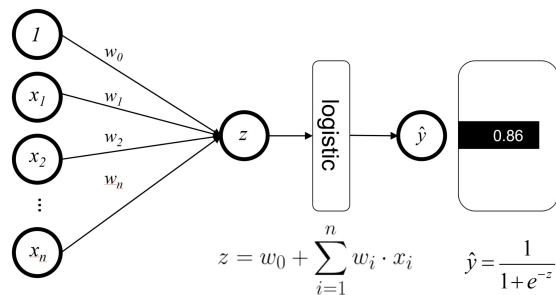
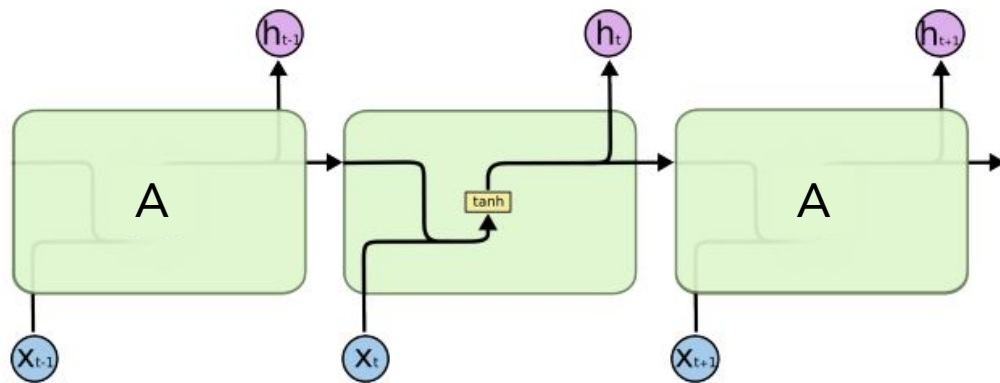
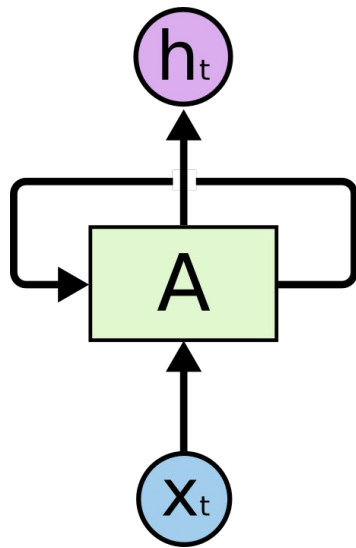
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Recurrent Neural Network

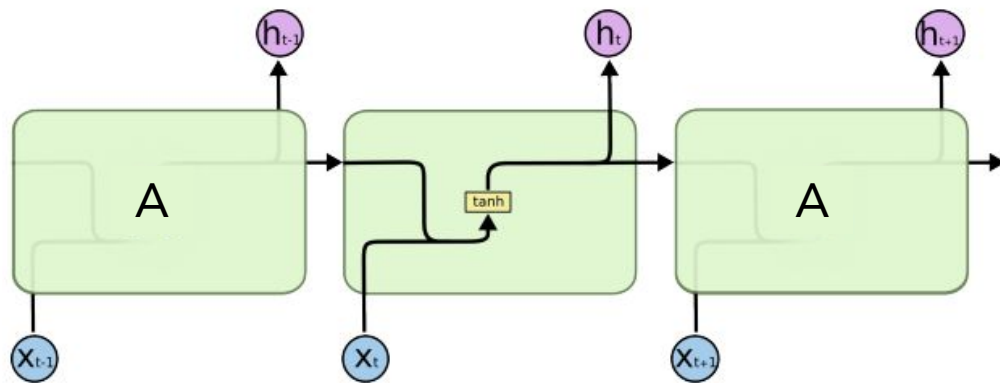
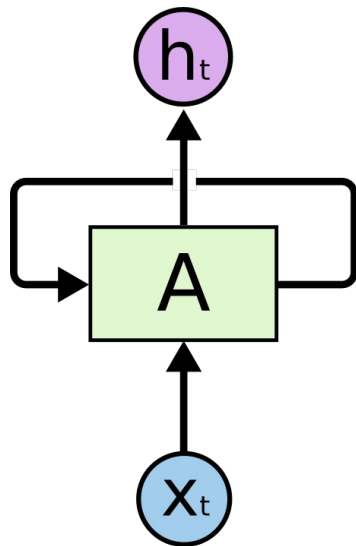


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Simple RNN Cell



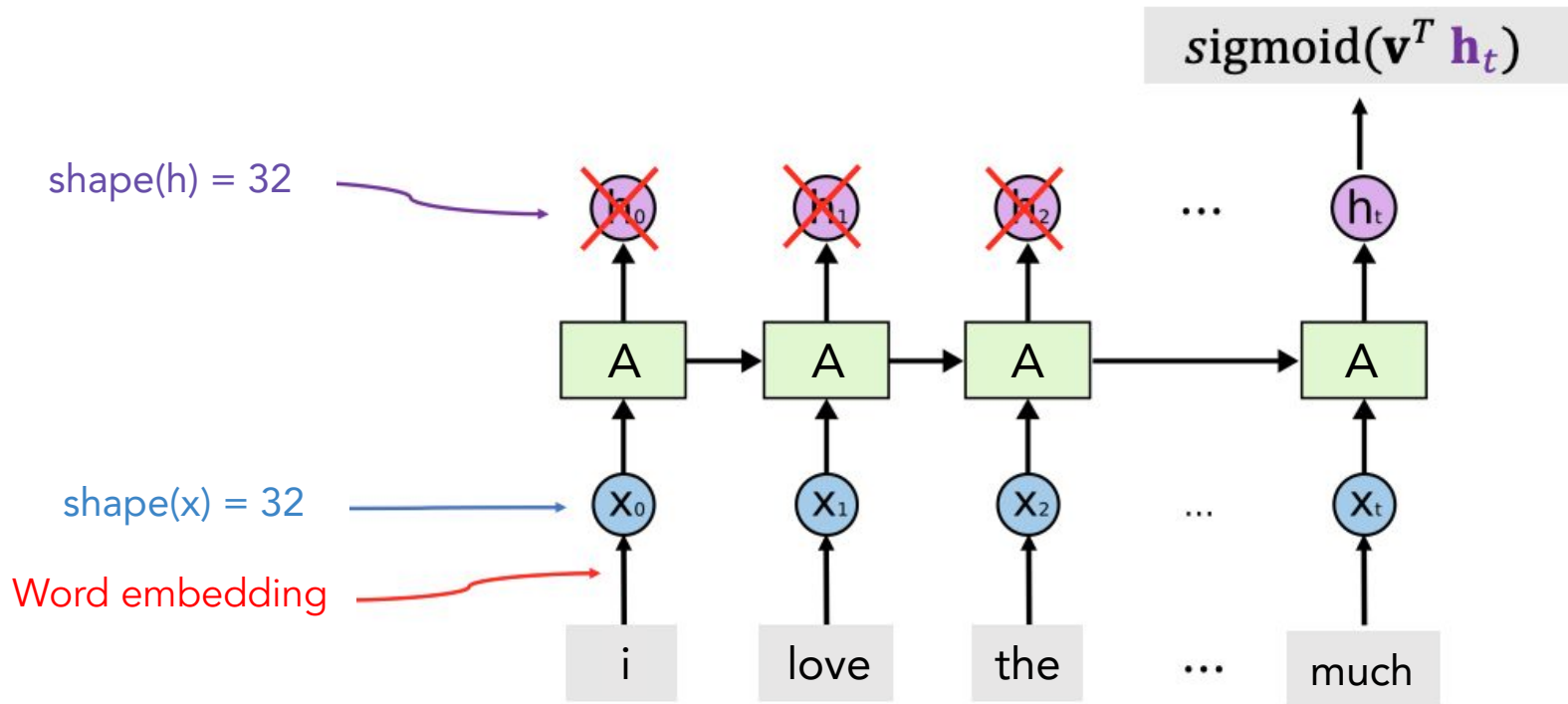
Simple RNN Cell



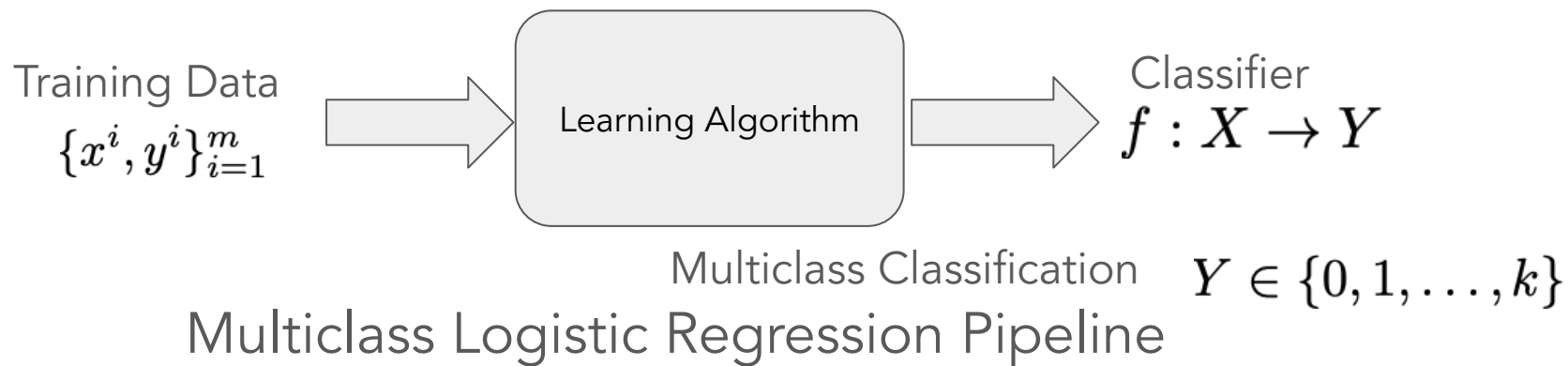
Binary step		$\begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$
Logistic, sigmoid, or soft step		$\sigma(x) \doteq \frac{1}{1 + e^{-x}}$
Hyperbolic tangent (tanh)		$\tanh(x) \doteq \frac{e^x - e^{-x}}{e^x + e^{-x}}$
Rectified linear unit (ReLU) ^[13]		$(x)^+ \doteq \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases} = \max(0, x) = x \mathbf{1}_{x>0}$
Gaussian Error Linear Unit (GELU) ^[5]		$\frac{1}{2}x \left(1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right) \text{ where erf is the gaussian error function.}$ $= x\Phi(x)$
Softplus ^[14]		$\ln(1 + e^x)$
Exponential linear unit (ELU) ^[15]		$\begin{cases} \alpha (e^x - 1) & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$ with parameter α

$$\mathbf{h}_t = \tanh \left[\mathbf{A} \cdot \mathbf{h}_{t-1} + \mathbf{x}_t \right]$$

Simple RNN for IMDB Review

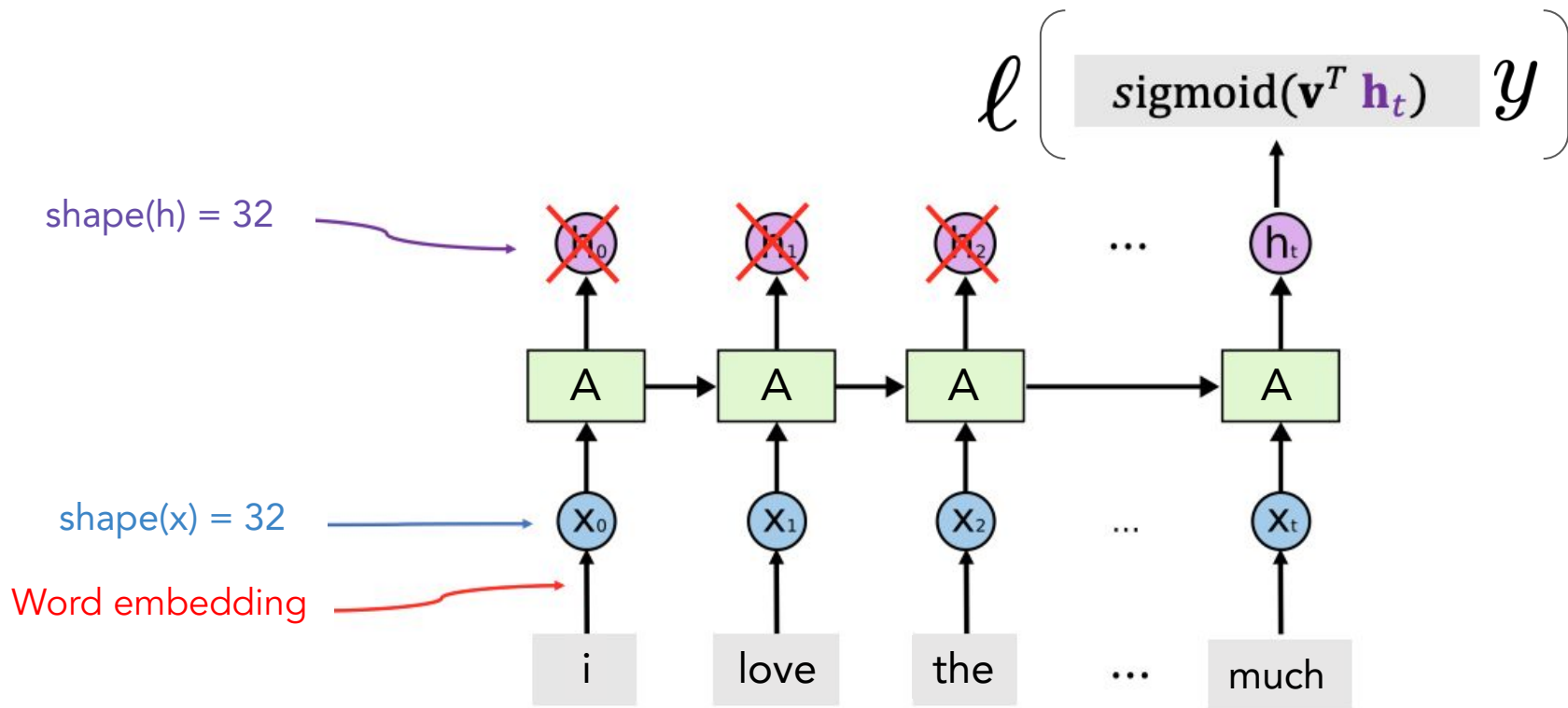


Sequential Multiclass Logistic Regression Algorithms



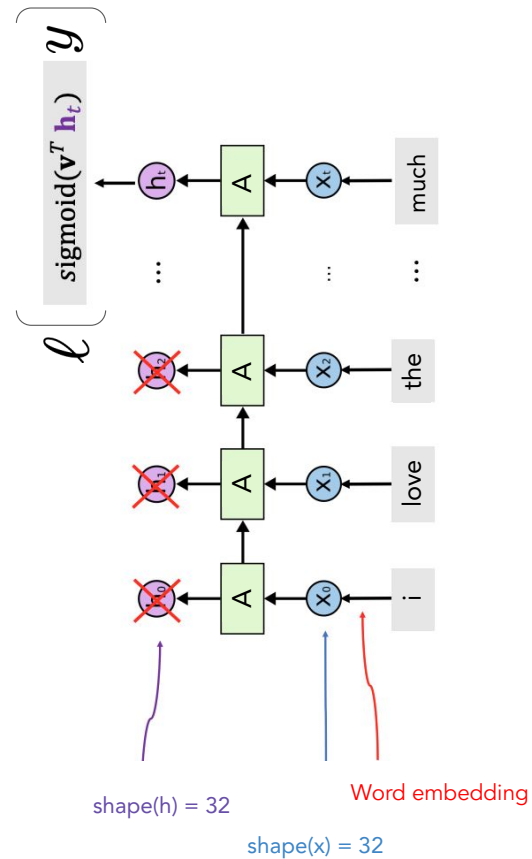
1. Build probabilistic models:
Categorical Distribution + RNN
2. Derive loss function: MLE and MAP
3. Select optimizer: (Stochastic) Gradient Descent

Backpropagation SGD



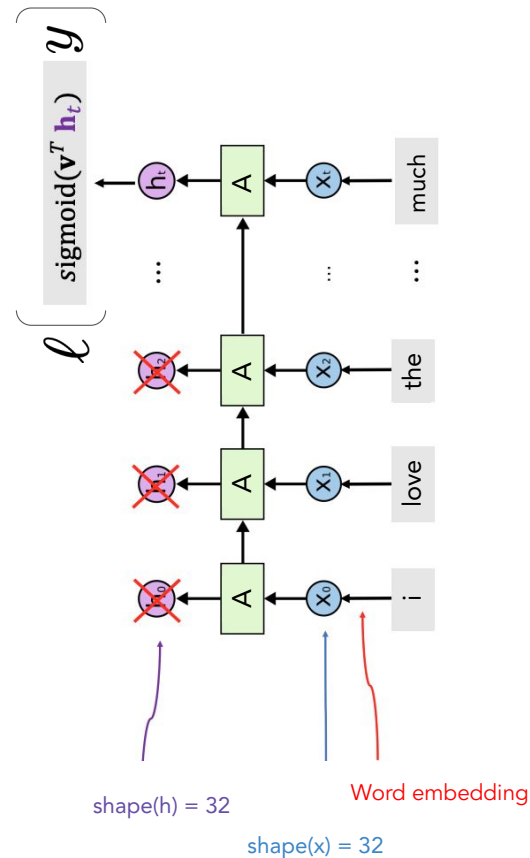
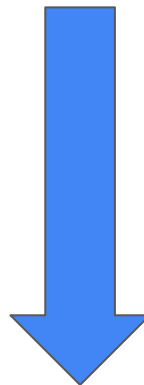
Backpropagation SGD

Layer $\longleftrightarrow$ Time-step

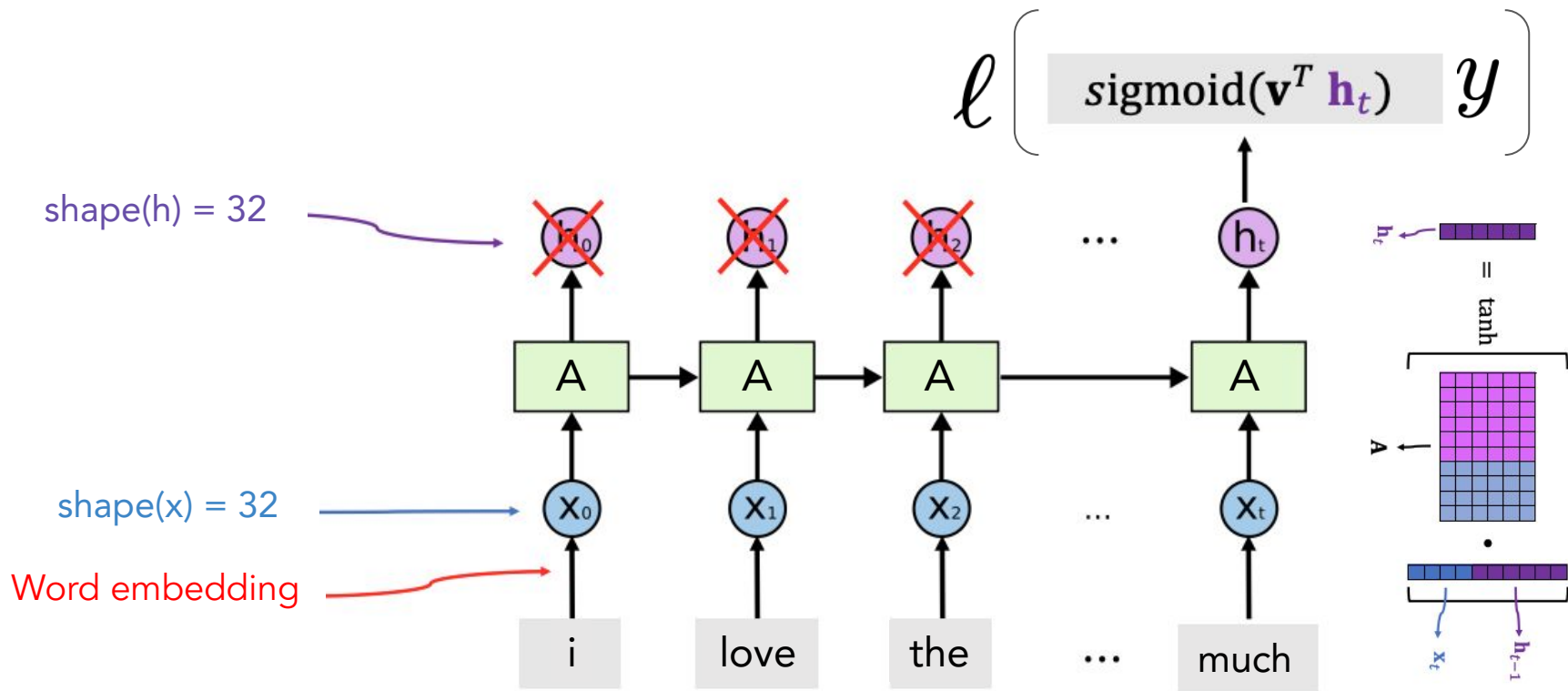


Backpropagation SGD

Backpropagation through Time
(BPTT)

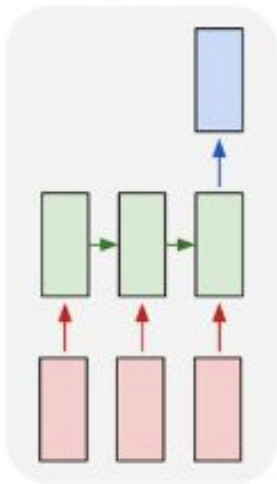


Backpropagation SGD



More Usages of RNNs

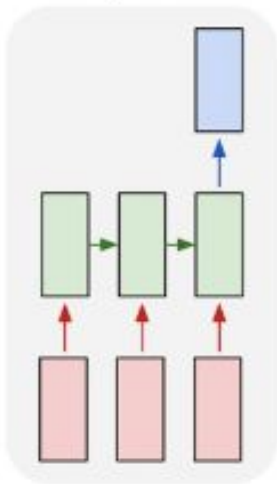
many to one



IMDB text review classification

More Usages of RNNs

many to one



one to many

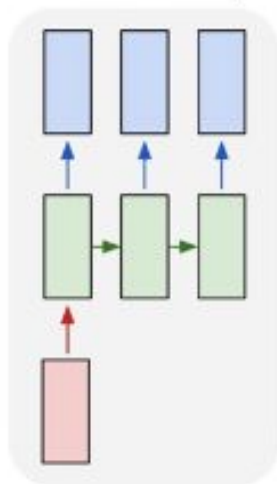
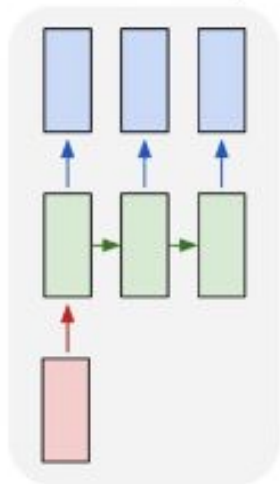


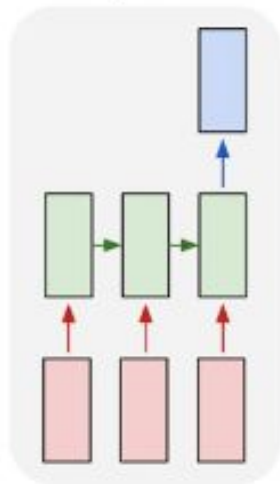
Image Captioning
image -> sequence of words

More Usages of RNNs

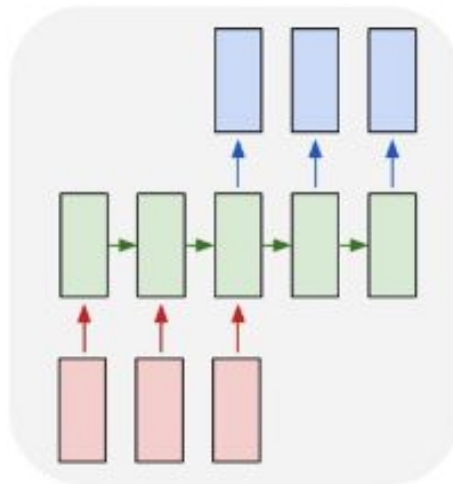
one to many



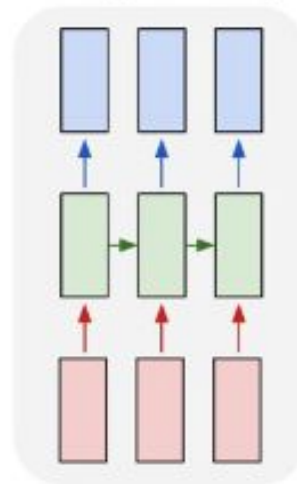
many to one



many to many

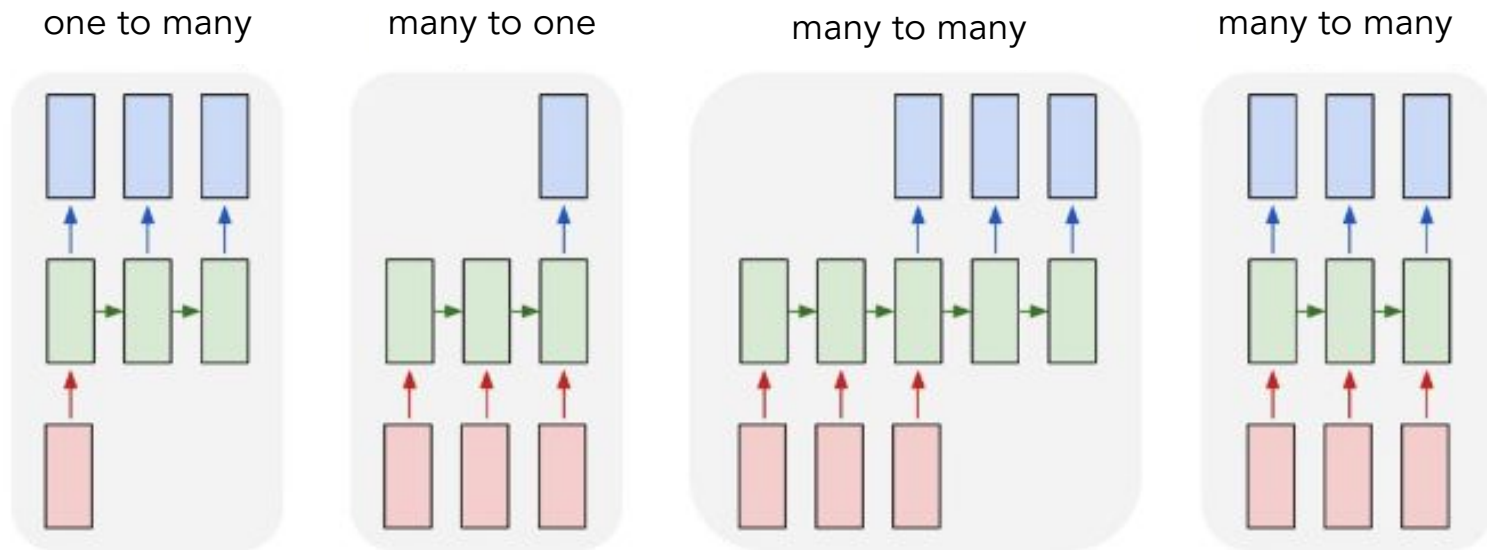


many to many



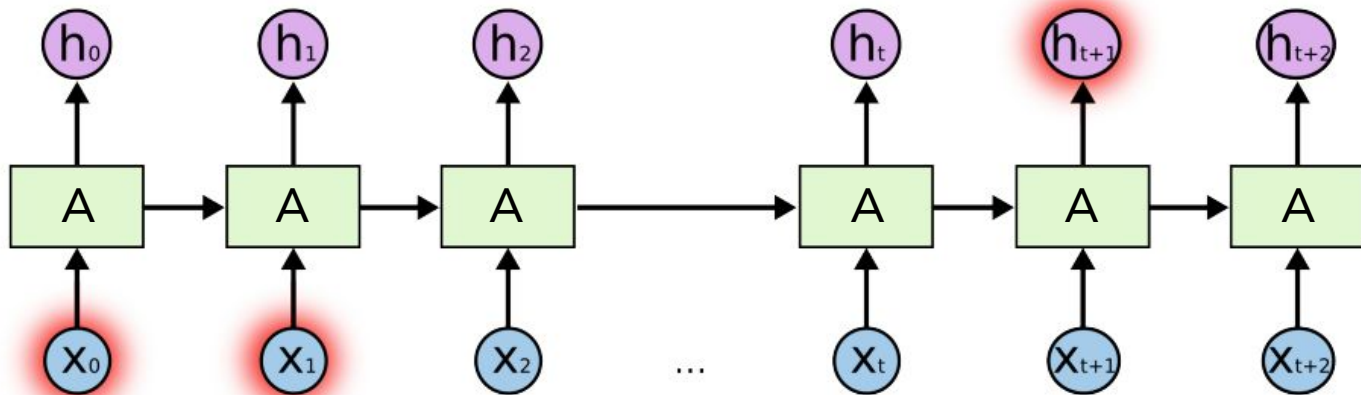
Translation

Training of RNNs



Backpropagation Through Unrolling Steps

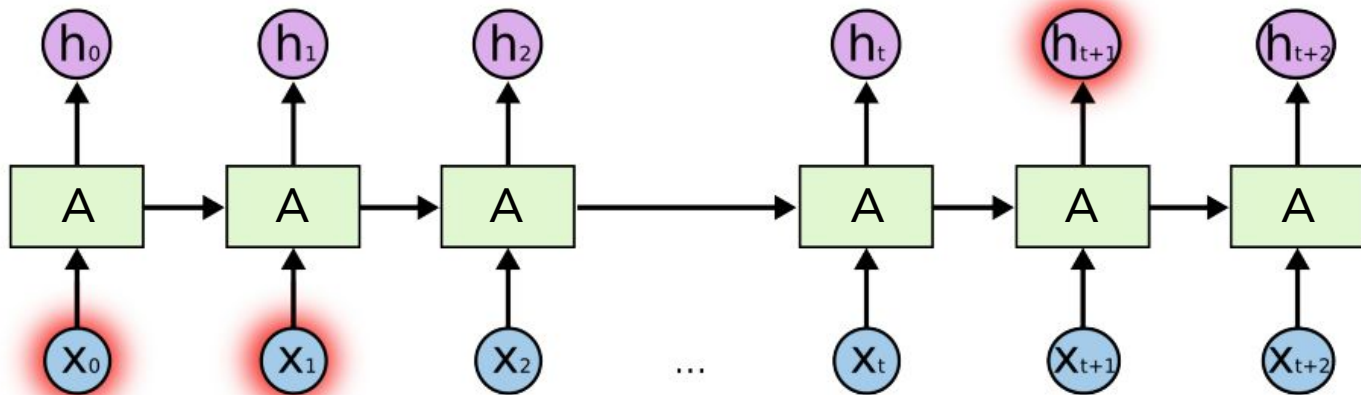
Simple RNN is not good at long-term dependence



$\mathbf{h}_{100}$ is almost irrelevant to $\mathbf{x}_1$: $\frac{\partial \mathbf{h}_{100}}{\partial \mathbf{x}_1}$ is near zero or exploding.

Gradient Vanishing Again!

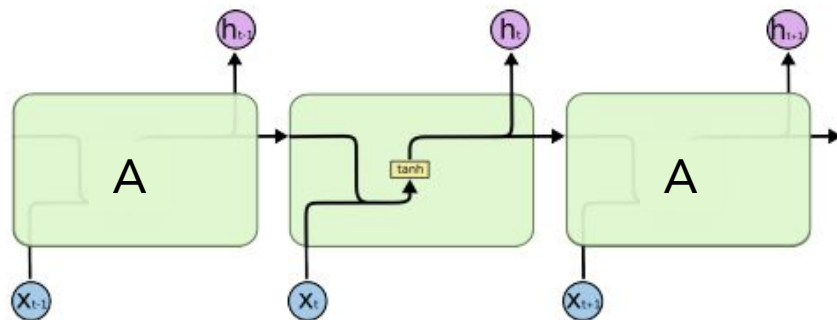
Simple RNN is not good at long-term dependence



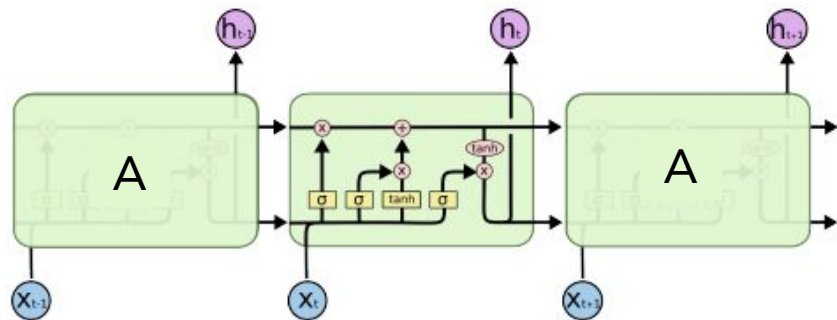
$\mathbf{h}_{100}$ is almost irrelevant to $\mathbf{x}_1$: $\frac{\partial \mathbf{h}_{100}}{\partial \mathbf{x}_1}$ is near zero or exploding.

Gradient Vanishing Again! -> ShortCut connection

Long Short Term Memory (LSTM)



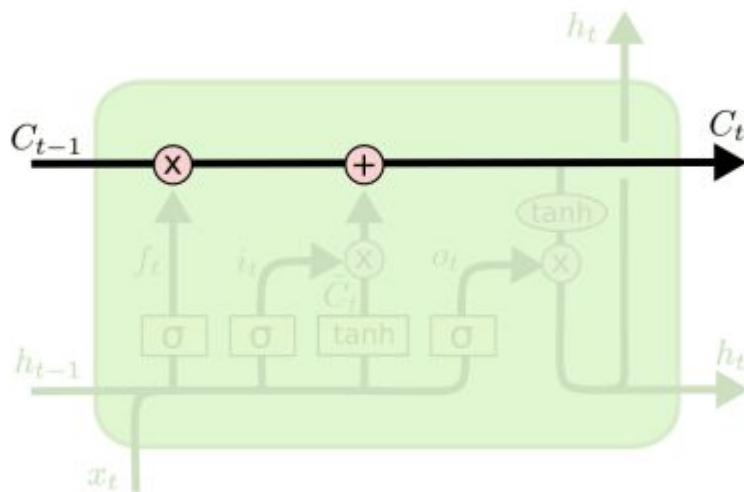
Simple RNN



LSTM

LSTM Cell: Conveyor Belt

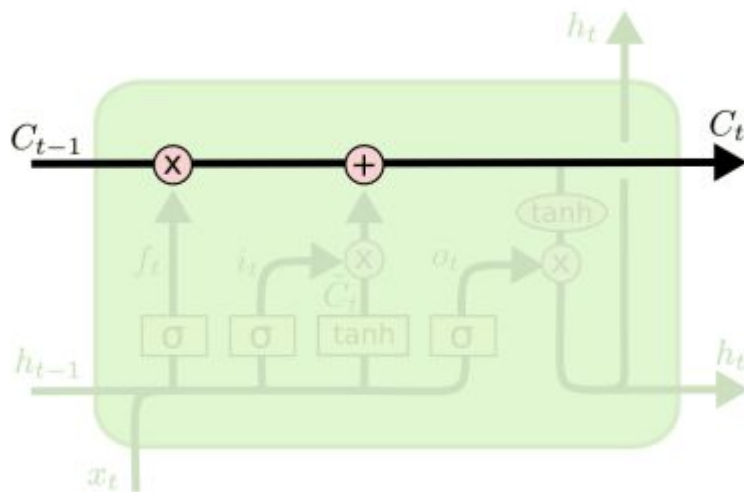
The past information directly flows to the future.



$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

LSTM Cell: Conveyor Belt

The past information directly flows to the future. ([ShortCut connection](#))

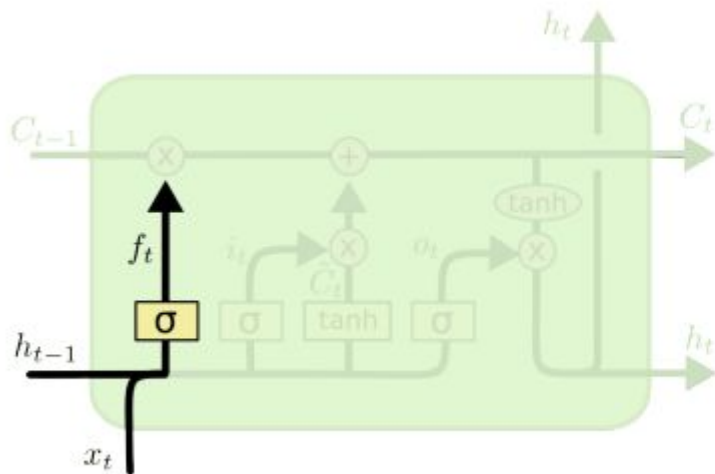


$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

LSTM Cell: Forget Gate

A value of zero means “let *nothing* through.”

A value of *one* means “let *everything* through!”



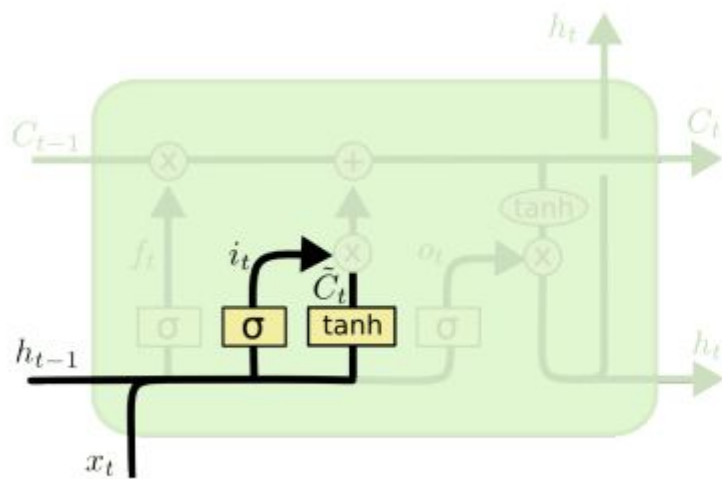
$$f_t = \sigma(W_f x_t + U_f h_{t-1})$$

The matrix representation of the forget gate equation is shown. A vertical vector f_t (purple) is equal to the sigmoid function σ applied to the sum of a weight matrix W_f (purple and blue grid) multiplied by the input vector x_t (blue) and a weight matrix U_f (purple grid) multiplied by the hidden state vector h_{t-1} (purple).

$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

LSTM Cell: Input Gate

How much information current context provided.



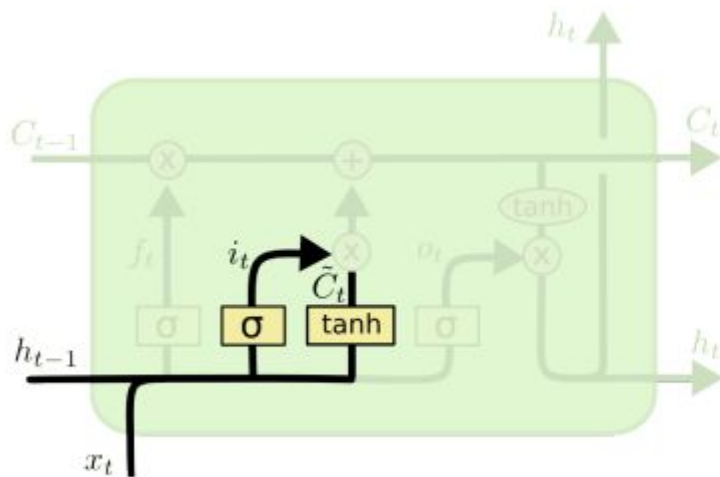
$$i_t = \sigma(W_i x_t + U_i h_{t-1})$$

The matrix representation of the input gate equation is shown. A vertical vector i_t (purple) is equal to the sigmoid function σ applied to the product of a weight matrix W_i (purple and blue grid) and a hidden state vector h_{t-1} (purple), plus the product of a weight matrix U_i (blue grid) and an input vector x_t (blue). The output vector i_t is shown as a vertical vector of purple and blue cells.

$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

LSTM Cell: New Value

“local” context, only up to immediately preceding state



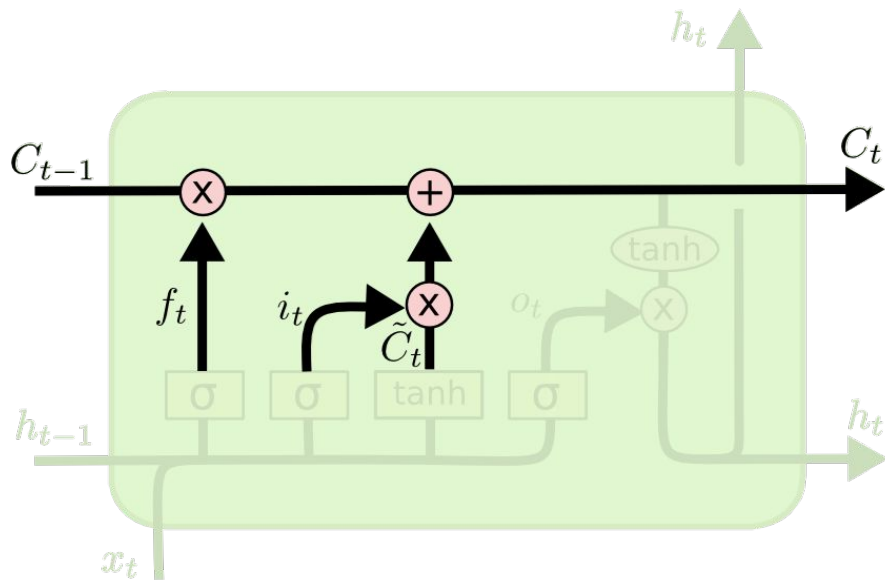
$$\hat{C}_t = \sigma(W_c x_t + U_c h_{t-1})$$

The diagram shows the matrix representation of the candidate cell state calculation. A vertical vector $\tilde{c}_t$ is equal to the $\tanh$ function applied to the product of a weight matrix W_c and a concatenated input vector. The input vector is formed by concatenating the previous hidden state h_{t-1} (purple) and the current input x_t (blue). The weight matrix W_c is shown as a grid of purple and blue squares, indicating its interaction with the concatenated input vector.

$$\tilde{c}_t = \tanh \left[W_c \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} \right]$$

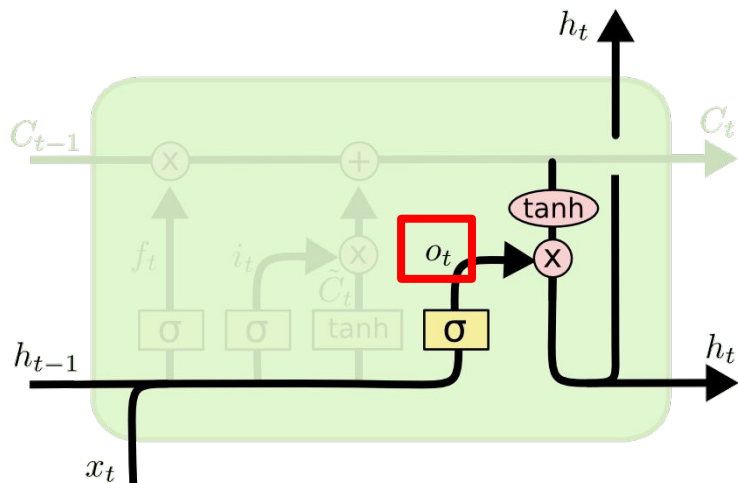
$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

LSTM Cell: Update Conveyor Belt



$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

LSTM Cell: Output Gate

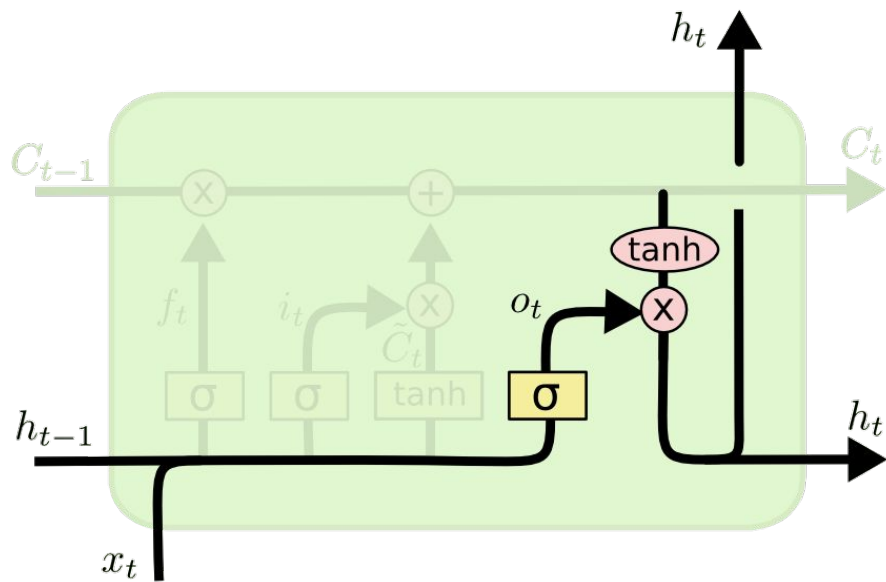


$$h_t = o_t * \tanh(C_t)$$

The equation shows the output gate output o_t as a vector of red squares. This is equal to the sigmoid function σ applied to the product of the weight matrix W_o (a grid of purple and blue squares) and the previous hidden state h_{t-1} (a vector of purple squares). The result is then multiplied by the current input x_t (a vector of blue squares).

$$o_t = \sigma \left[W_o \cdot \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} \right]$$

LSTM Cell: Update State



A vector-based representation of the state update equation. On the left, a purple vertical vector represents the hidden state h_t . This is equal to a red vertical vector representing the output gate output o_t , followed by an element-wise multiplication ($\circ$) with the tanh of a green vertical vector representing the cell state C_t . The green vector C_t is enclosed in square brackets, and a green arrow points from it to the label C_t below.

$$h_t = o_t * \tanh(C_t)$$

Auto-differentiation Packages

PyTorch



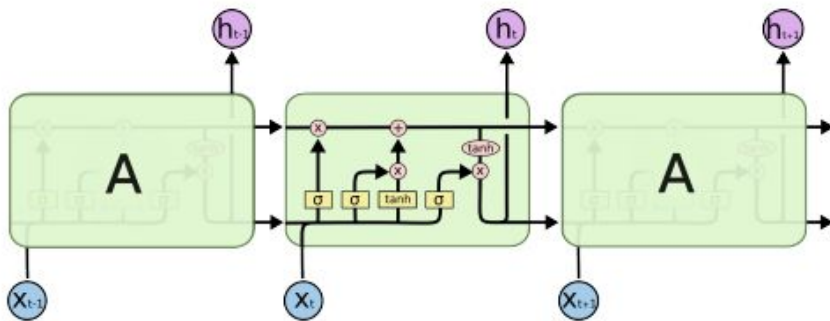
JAX



Tensorflow

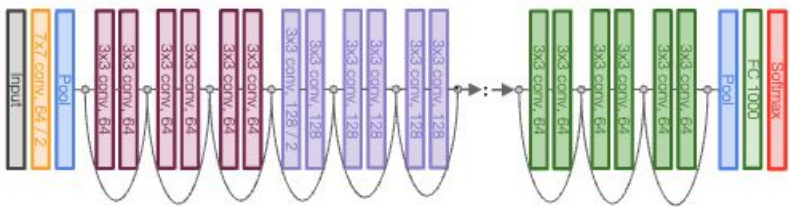


LSTM vs. ResNet



$$C_t = C_{t-1} \circ f_t + \hat{C}_t \circ i_t$$

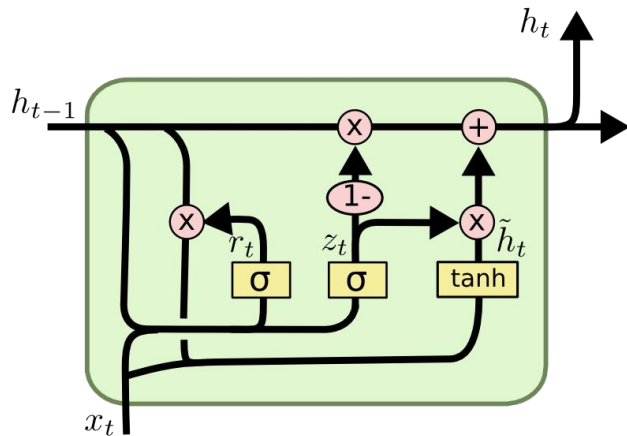
LSTM



Similar to ResNet

Other Variants of RNN

Gated Recurrent Unit



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

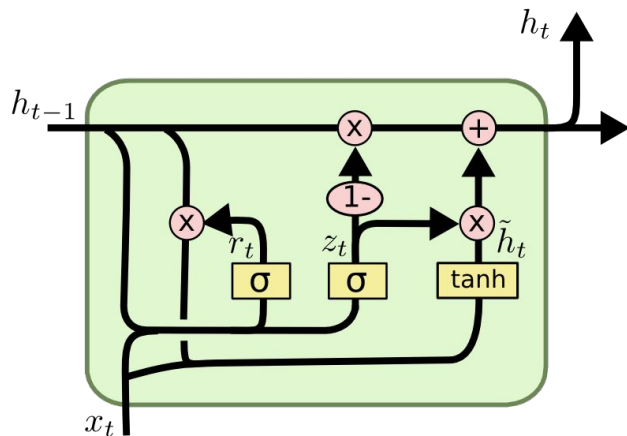
$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

Other Variants of RNN

Gated Recurrent Unit



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

Summary

- RNNs allow a lot of flexibility in architecture design

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- RNNs allow a lot of flexibility in architecture design
- Vanilla RNNs are simple but don't work very well
- Backward flow of gradients in RNN can explode or vanish. Exploding is controlled with gradient clipping. Vanishing is controlled with additive interactions (LSTM)

Q&A